

Question 1 : Describe different types of data sources used in ETL with suitable examples.

Here are the types of data sources used in ETL in short points:

- **Relational Databases (Structured Data)** - Data stored in tables (rows & columns). Examples: MySQL, Oracle Database
- **Flat Files** - Text-based files like CSV, Excel .Example: Microsoft Excel
- **NoSQL Databases** - Semi-structured or unstructured data. Examples: MongoDB, Apache Cassandra
- **Cloud Applications / SaaS** - Online platforms providing data via cloud .Examples: Salesforce, Google Analytics
- **Web Services / APIs** - Data accessed through REST or SOAP APIs. Example: Twitter API
- **Data Warehouses** - Centralized storage systems. Examples: Amazon Redshift, Snowflake
- **Big Data Platforms** - Large-scale distributed data systems. Examples: Apache Hadoop, Apache Spark
- **Streaming Data Sources** - Real-time continuous data. Example: Apache Kafka

Question 2 : What is data extraction? Explain its role in the ETL pipeline.

Data extraction simply means collecting data from different sources so it can be used somewhere else, like in a data warehouse or reporting system.

Role of Data Extraction in the ETL Process

- **It is the first step** – ETL starts with extracting data.
- **It gathers data from different places** – like databases (e.g., MySQL), spreadsheets (e.g., Microsoft Excel), or online tools (e.g., Salesforce).
- **It brings all data together** – so it can be cleaned and organized later.
- **It makes analysis possible** – without extracting data, we cannot transform it or load it for reporting.

Question 3 : Explain the difference between CSV and Excel in terms of extraction and ETL usage.

Feature	CSV	Excel
File Type	Simple text file	Formatted spreadsheet file
Complexity	Very simple	More complex
Extraction Speed	Faster	Slightly slower
Supports Formatting	No	Yes
Multiple Sheets	No	Yes
Best Used For	Data transfer	Reports and analysis

Question 4 : Explain the steps involved in extracting data from a relational database

Here are the steps :

- **Choose the database** - Decide which database you want data from (for example, MySQL or Oracle Database).
- **Connect to the database** - Use a username and password to log in.
- **Pick the data you need** - Select the specific tables and columns (like customer name, sales amount, date).
- **Run a query (SQL command)** - Write a simple SQL command to get the required data.
- **Choose how much data to take** - Take all data (full extraction), or take only new/updated data (incremental extraction).
- **Save it in a temporary place** - Store the extracted data in a staging area before cleaning or transforming it.
- **Check the data** - Make sure there are no missing or duplicate records.

Question 5 : Explain three common challenges faced during data extraction.

Three Common Problems During Data Extraction -

- **Bad or Messy Data**
Sometimes the data is Missing information ,Repeated (duplicates) ,Incorrect or wrongly typed .For example, the same customer name might be written in different ways in a database like MySQL. This makes reports and analysis wrong or confusing.

- **Different File Formats**

Data can come from different places like: Databases ,CSV files ,Excel sheets (like Microsoft Excel) Each source may store dates, numbers, or text differently. Because of this, combining the data becomes difficult.

- **Too Much Data**

Sometimes there is a very large amount of data. This can Slow down the extraction process ,Use more storage space ,Affect system performance.

Question 6 : What are APIs? Explain how APIs help in real-time data extraction.

API stands for Application Programming Interface. An API is a bridge that allows two software systems to talk to each other and share data.

For example, when one application wants data from another system, it sends a request through an API, and the other system sends the data back.

Many platforms provide APIs, such as Twitter and Salesforce.

APIs are very useful in ETL, especially for real-time data. Here's how:

- **Instant Data Access**

APIs allow systems to request and receive data immediately.

- **Automatic Data Updates**

Data can be fetched automatically at regular intervals without manual work.

- **Live Data Streaming**

Some APIs provide continuous data updates (for example, live transactions or social media posts).

- **Easy Integration Between Systems**

APIs make it easier to connect different applications and transfer data smoothly.

Question 7 : Why are databases preferred for enterprise-level data extraction?

For big companies, databases are usually the go-to choice for extracting data because they make the process faster, safer, and easier. Here's why:

- **Organized Data** - Data is stored neatly in tables (rows and columns), so it's easy to find what you need. Example: MySQL or Oracle Database.

- **Can Handle Huge Amounts of Data** - Databases can manage millions of records without slowing down.

- **Reliable and Accurate** - They follow rules that prevent mistakes, duplicates, or missing data.

- **Smart Data Extraction** - You can extract only new or updated data instead of pulling everything every time.
- **Secure Access** - Only authorized users or ETL tools can access sensitive information.

Question 8 : What steps should an ETL developer take when extracting data from large CSV files (1GB+)?

When a CSV file is really big, like 1GB or more, here's what an ETL developer does to handle it safely:

- **Check the File First** - Make sure it's not broken and the data looks okay.
- **Use a Temporary Place (Staging Area)** - Don't load it directly into the main system. Keep it somewhere safe first.
- **Read the File in Small Parts** - Instead of opening the whole file at once, read it in smaller chunks so the computer doesn't crash.
- **Take Only What You Need** - Extract only the rows and columns needed to make it faster.
- **Use Tools That Can Handle Big Files** - Use ETL tools or programming libraries that are made for large data.
- **Watch System Performance** - Keep an eye on memory and storage while extracting.
- **Check the Data After Extraction** - Make sure nothing is missing or broken before moving to the next ETL step.